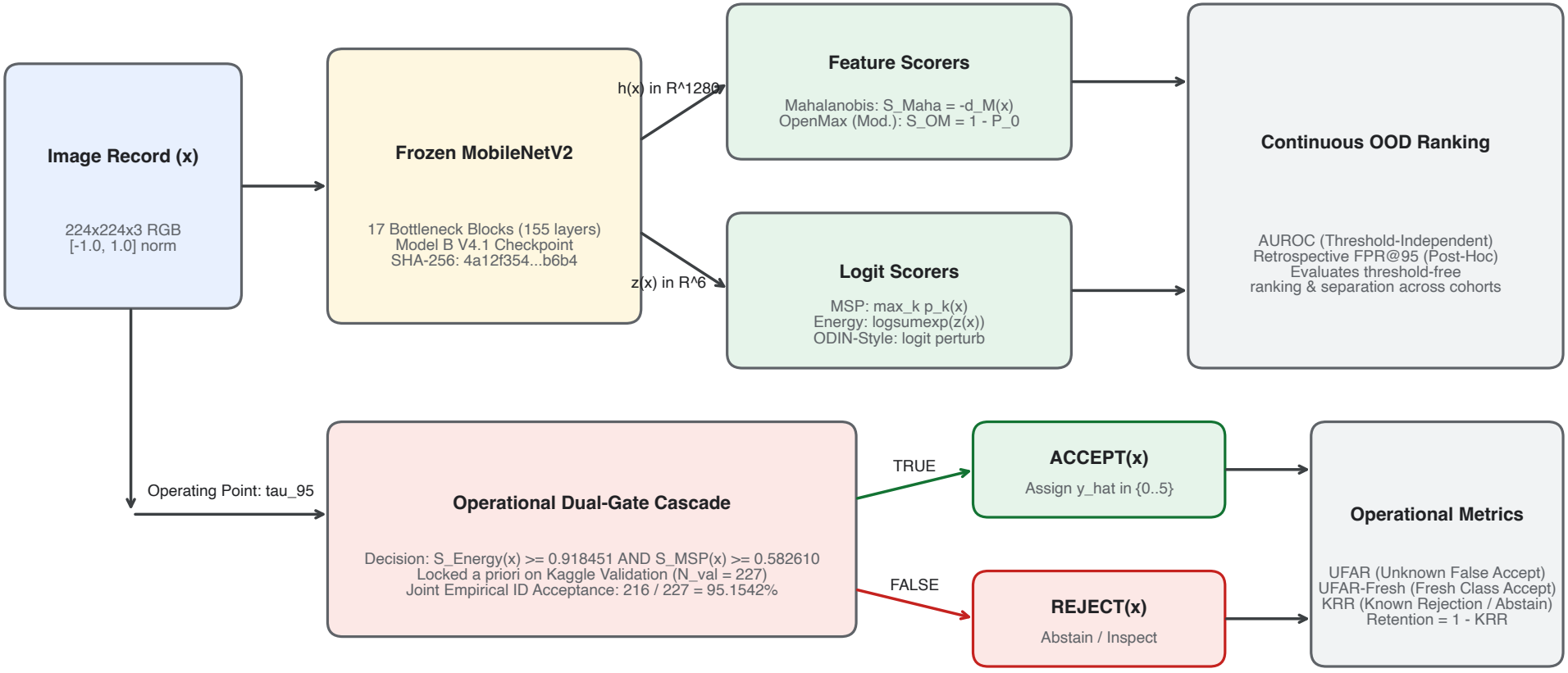
Image Record (x)
224x224x3 RGB
[-1.0, 1.0] norm
Frozen MobileNetV2
17 Bottleneck Blocks (155 layers)
Model B V4.1 Checkpoint
SHA-256: 4a12f354...b6b4
h(x) in R^1280
z(x) in R^6
Feature Scorers
Mahalanobis: S_Maha = -d_M(x)
OpenMax (Mod.): S_OM = 1 - P_0
Logit Scorers
MSP: max_k p_k(x)
Energy: logsumexp(z(x))
ODIN-Style: logit perturb
Continuous OOD Ranking
AUROC (Threshold-Independent)
Retrospective FPR@95 (Post-Hoc)
Evaluates threshold-free
ranking & separation across cohorts
Operating Point: tau_95
Operational Dual-Gate Cascade
Decision: S_Energy(x) >= 0.918451 AND S_MSP(x) >= 0.582610
Locked a priori on Kaggle Validation (N_val = 227)
Joint Empirical ID Acceptance: 216 / 227 = 95.1542%
TRUE
FALSE
ACCEPT(x)
Assign y_hat in {0..5}
REJECT(x)
Abstain / Inspect
Operational Metrics
UFAR (Unknown False Accept)
UFAR-Fresh (Fresh Class Accept)
KRR (Known Rejection / Abstain)
Retention = 1 - KRR